

1 Owner-verdict response: the narrow win and a bounded screen

The central computational result is an aligned aggregation comparison on pep424, not a three-task victory. The committed `results/pep424_v3.json` has 419 aligned examples: transferred pepx-GNN AUROC 0.8391 and FoldAmyloid predictions 0.7480 on those rows, a difference of 0.0911. The internal five-fold pepx-GNN AUROC 0.7947 is below AmyloGram’s separately published CV 0.865, which is a different comparison. AntiCP 2.0 main and alternate published bars remain unbeaten, including the two subsequently locked official-test attempts (0.7884 and 0.8121 AUROC against the 0.83 main benchmark). The claim should be *one same-row comparator advantage under the named protocol*, not a general superiority over peptide methods.

1.1 Data alignment and integrity are part of the scientific result

A name-based pep424 join was invalid because a generic header name covered many rows. The corrected v3 comparison aligns the evaluation and published prediction order, excludes noncanonical sequences and uses its recorded 419 examples. The sequence and score alignment are necessary conditions for the comparator claim; they do not create an independent training split or prove transfer to a new aggregation cohort. AntiCP 2.0 test peptides include near-homologs to training, and strong composition descriptors limit what a learned architecture can be credited with. A five-fold average without multiple model seeds does not measure seed variance.

The molecular-weight table passed a narrow A/K/L lock while P/Q and V/W/Y were wrong. A third independent validator exposed the errors, and a full-residue regression test now covers the corrected table. The repair changed the pep424 transfer AUROC from the previously invalid 0.7902 to the committed 0.8391; earlier numbers must not be compared as though only the model had improved. Separately, the 25-epoch scaled-screen checkpoint was nearly written over the shipped `trinet.pt`. The ONNX parity check exposed changed scoring; the old shipped weights were restored from git before committing the drifted checkpoint under a distinct filename. No claim that the drifted model was shipped is made. These incidents delimit the provenance of the usable artifact, not evidence for peptide activity.

1.2 Constrained screening, not discovery

The tri-objective score multiplies model outputs for anticancer, solubility and low aggregation. That objective and the hand-set charge/hydrophathy gates define a search procedure. They are not calibrated probabilities of clinical success. The initial constrained candidate and later anionic drift from longer training show that model selection and screen thresholds can move the searched chemical space. A high score from the same model that guided annealing is not an independent assessment. The candidate set is limited to the searched 10-30-residue regime, and neither membrane structure nor disulfide topology is resolved by the current representation. No cancer-cell selectivity, hemolysis, bioavailability or toxicity experiment validates a nomination.

A future screen must lock its checkpoint hash, source-database versions, allowed mutations, charge/hydrophathy bounds, novelty metric, toxicity and disulfide handling, and assay plan before scoring new candidates. It should report all excluded sequences and failures, use an independent predictor as a stress test, and keep experimental verification separate from score ranking.

HemoPI-class hemolysis modeling, ESM-2/ProtT5 embeddings, per-residue solubility, disulfide-aware gating, BLAST novelty, and a 60% test-coverage target are [PENDING]; none is an observed improvement. The strict tools ledger currently lists 24 candidate research/data tools, not a verified 40. Accession records are not independent cohorts. This response adds methods interpretation and limitations, not a slide presentation.